



Rapid Estimation of Truck Cycle Time in Open-Pit Mine Haulage Based on Feature-Optimized Machine Learning

Chengkai Fan^{1,3} · Na Zhang² · Bei Jiang² · Wei Victor Liu¹

Received: 30 October 2024 / Accepted: 26 February 2025 / Published online: 18 March 2025
© Society for Mining, Metallurgy & Exploration Inc. 2025

Abstract

Machine learning has received extensive attention in building prediction models of truck cycle time based on mine haulage. However, the data collected from mine sites are usually massive and contain numerous inputs. Machine learning is known for handling massive data, establishing complex input–output relationships, and providing fast computations. Thus, three machine learning algorithms, including extreme learning machine (ELM), extremely randomized trees (ERT), and extreme gradient boosting (XGBoost), were employed to train prediction models of mining truck cycle time. To further decrease computational costs and enhance the model accuracy, this study conducted feature optimization by using three feature selection methods and two unsupervised clustering techniques. The results showed that XGBoost outperformed ELM and ERT in predicting truck cycle time. Recursive feature elimination (RFE) was the best feature optimization approach to remove redundant variables and enhance the model performance. Among the selected inputs, haul distance, empty speed, and waiting at shovel were the most critical variables affecting truck cycle time. In addition, Gaussian mixture modeling (GMM) significantly improved the model accuracy, but K-means could not increase the model predictability. The hybrid models (e.g., GMM-RFE-ELM) had high prediction accuracy with the shortest computational time. This study developed three extreme hybrid models to accurately and rapidly estimate mining truck cycle time, which helps mining engineers make better decisions.

Keywords Machine learning · Feature optimization · Truck haulage · Clustering techniques · Prediction performance

1 Introduction

Oil sands mining contributes immensely to the national economic output of Canada [5]. The GDP of oil sands mining has been projected at CAD\$ 2106 billion for Canada over

25 years (2010–2035) [28]. In oil sands mining, off-the-road trucks play a leading role in transporting bulk materials (ores and waste) at open-pit mine sites [38]. The time required for a truck to complete a haulage cycle is referred to as truck cycle time, consisting of time for loading, hauling, dumping, returning, and waiting [54]. Mining truck cycle time is of great interest to the resource industry since it is a critical indicator in assessing the maximum productivity achievable between load and dump sites, which directly affects a mine's overall productivity, production targets, planning, and budgets [12, 13].

To estimate truck cycle time, machine learning has attracted considerable attention because of its ability to build data-driven prediction models [22, 24]. Machine learning is a general term for a set of mathematical algorithms that can automatically acquire information from historical data and establish input–output relationships (i.e., prediction models) [2]. These algorithms have been successfully employed for various mining applications, such as rock-burst prediction [66], cement materials [63], mining safety [56], and waste management [16]. Among these algorithms,

✉ Bei Jiang
bei1@ualberta.ca

✉ Wei Victor Liu
victor.liu@ualberta.ca

Chengkai Fan
chengkai.fan@gmn.ulaval.ca

Na Zhang
nz4@ualberta.ca

¹ School of Mining and Petroleum Engineering, University of Alberta, Edmonton, AB T6G 2G1, Canada

² Department of Mathematical and Statistical Sciences, University of Alberta, Edmonton, AB T6G 2G1, Canada

³ Department of Mining, Metallurgical and Materials Engineering, Laval University, Quebec City, QC G1V 0A6, Canada

extreme learning machine (ELM), extremely randomized trees (Extra-trees or ERT), and extreme gradient boosting (XGBoost) are well-known machine learning methods for handling massive amounts of data, which are superior in prediction performance and time efficiency [24, 62]. For example, Dhini et al. [18] proposed an ELM model for fault diagnosis of steam turbines in thermal power plants. The research presented that ELM achieved sound prediction accuracy, with a coefficient of determination (R^2) of 96.58%. Meanwhile, the computational time of this model was 54.81 s, which performed significantly faster than the widely used backpropagation neural network (BPNN) model (161.12 s). Likewise, both Chencho et al. [15] and Saeed et al. [48] built ERT and random forest (RF) models for two-element damage quantification of concrete structures and fault detection in wireless sensor networks, respectively. According to Chencho et al. [15], the accuracy of the ERT model (98.5%) was better than that of the RF model (97.5%), the computational time of ERT (22.70 s) was considerably shorter than the latter (101.59 s). Saeed et al. [48] also observed that ERT (81.20%) was superior to RF (79.60%) and reduced the running time by more than half, from 190 to 90 s. Moreover, Demir and Sahin [17] used XGBoost and two traditional boosting methods, including gradient boosting regression (GBR) and adaptive boosting (AdaBoost), to forecast soil liquefaction in earthquake engineering. Compared with GBR and AdaBoost, XGBoost demonstrated the highest prediction accuracy (96.75%) and the smallest computational cost (5.15 s). Therefore, these machine learning algorithms have substantial potential to build accurate prediction models and provide fast computation for mining truck cycle time.

Nevertheless, no research has been found that uses these machine learning algorithms to estimate truck cycle time in open-pit mine haulage based on the current literature review. These extreme algorithms may bring vast benefits to predicting truck cycle time, which entails large amounts of real-site datasets and complex relationships between numerous inputs and outputs [21, 22]. For instance, the dataset in this study contained over 8600 data points encompassing a full year's truck haulage information. Moreover, many input variables are available at mine sites according to the site observation [24, 54], such as haul distance, time-related factors, truck running speeds, haul routes, and meteorological conditions (e.g., ambient temperature and precipitation). These usable data and variables may cause a dramatic growth in computation costs and high nonlinearity of prediction models [21]. Extreme algorithms can overcome these prominent problems, yet this potential remains unknown.

To fill this research gap, the purpose of this study was to provide accurate and rapid estimations of truck cycle time in open-pit mine haulage by developing extreme machine learning models based on massive truck haulage data. Unlike previous studies that directly constructed prediction models

[2, 13], this study conducted two prior comparative studies to optimize the subset of input features: (1) comparing feature selection methods and (2) comparing unsupervised clustering techniques. First, three feature selection approaches were utilized to remove irrelevant input variables, including decision tree (DT), analysis of variance (ANOVA), and recursive feature elimination (RFE). These methods can remove irrelevant information and noise, thus decreasing computational costs and improving prediction performance [34]. Next, two unsupervised clustering techniques, including K-means and Gaussian mixture modeling (GMM), were applied to preprocess the datasets with the selected inputs, as these two techniques have been shown to effectively deal with massive data and enhance the model predictability [22, 36]. Finally, the ELM, ERT, and XGBoost models of truck cycle time were built based on feature optimization.

The novelty of this study stems from four aspects. First, an insightful analysis was carried out on unique and massive truck haulage data at open-pit mines. Second, three feature selection methods were compared and applied to analyze the real-world factors affecting mining truck cycle time. Third, ELM, ERT, and XGBoost were used to provide rapid truck cycle time estimations for the first time. Fourth, the ELM, ERT, and XGBoost models were combined with K-means and GMM to investigate the influence of clustering techniques on models' accuracy. This paper contributes to the development of feature-optimized extreme hybrid models to estimate mining truck cycle time accurately and rapidly.

The remainder of the manuscript is assembled below. Section 2 describes the overview framework of this research, data description, feature selection approaches (i.e., RFE, ANOVA, and DT), clustering techniques (i.e., K-means and GMM), and extreme machine learning algorithms (i.e., ELM, ERT, and XGBoost). Section 3 shows the comparative studies of feature selection methods and clustering techniques, compares the extreme prediction models' performance, assesses the impact of clustering techniques on the models' accuracy, and analyzes the models' computational efficiency. Lastly, Sect. 4 gives the main findings.

2 Methodology and Data Preparation

2.1 Research Framework

Figure 1 demonstrates the overall research framework. In Fig. 1, the data from truck haulage in oil sands mine sites were tabulated and separated into two datasets: training and testing. To remove irrelevant information in these datasets, three feature selection approaches (DT, ANOVA, and RFE) were first applied to optimize the essential subset of inputs. Next, the best-selected subset was classified by K-means and GMM to identify subgroups from massive

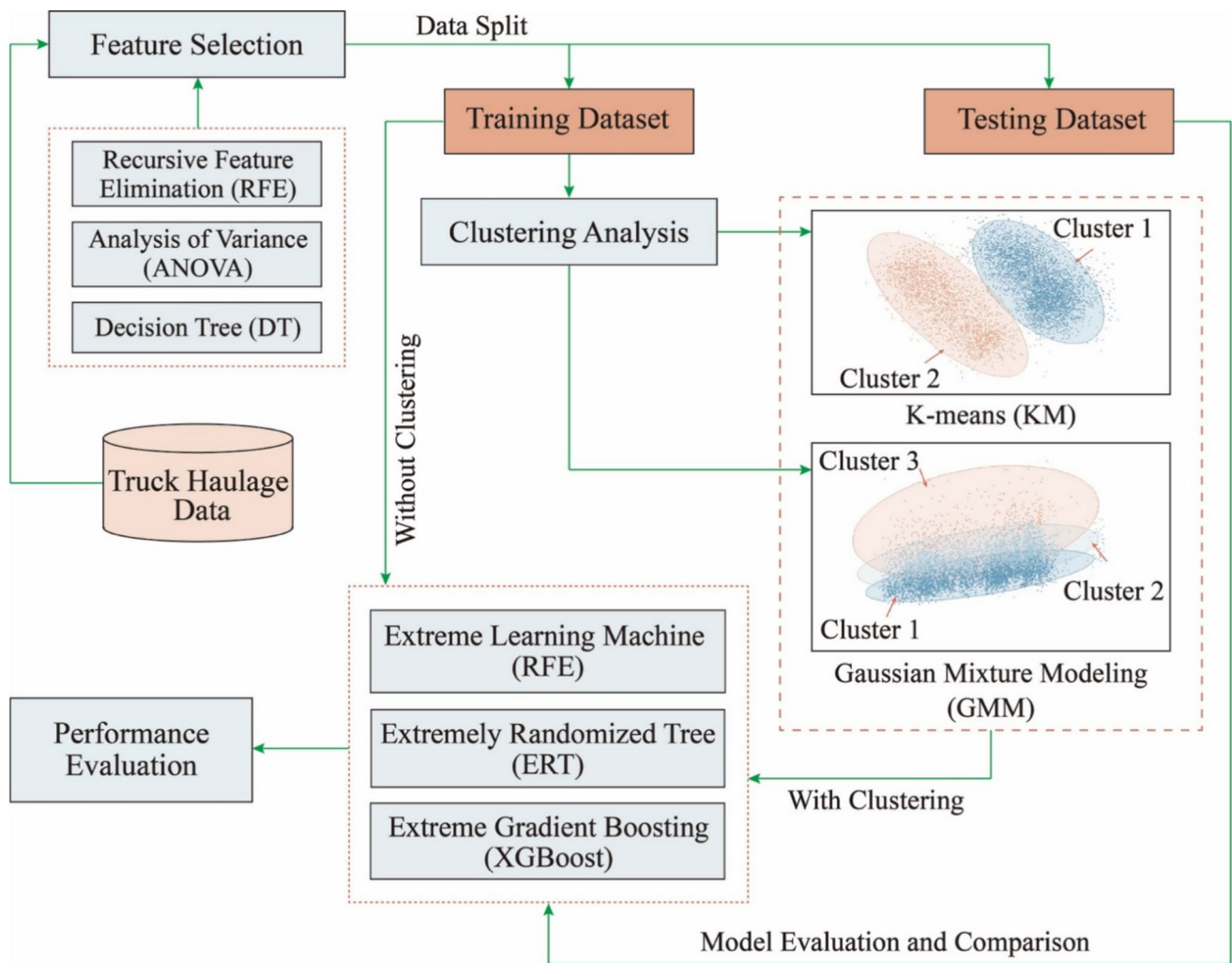


Fig. 1 Proposed study framework for extreme machine learning models developed in this study

data, respectively. Meanwhile, the labels of these subgroups constituted additional variables and were treated as new input variables (i.e., new categorical variables) incorporated with the selected subset to become new training datasets [21]. The testing dataset was correspondingly classified based on K-means and GMM analysis. After that, ELM, ERT, and XGBoost were adopted to develop prediction models using the new training datasets and the initially optimized subset, respectively. Finally, the testing dataset was utilized to assess the prediction performance. Four performance metrics were recommended [4]: the root mean square error (RMSE), the mean absolute error (MAE), R^2 , and running time. Overall, the training procedure was conducted on a personal computer (PC) using R programming (version 4.1.3) in RStudio software. This PC has a 64-bit operating system with 16.0 GB of RAM and an Intel Core i7-12700K (3.60 GHz) processor.

2.2 Data Description

The massive dataset was obtained from practical mine haulage roads in Alberta, Canada. It had 8683 groups of data points, each representing a truck haulage cycle recorded for each hour of the year. The dataset was partitioned into training (80%) and testing (20%) datasets. This split proportion was determined and recommended based on common practice in previous studies [2]. Both these two datasets included an output variable (cycle time, y) and ten input variables (x_m). These inputs were haul distance (x_1 , km), empty speed (x_2 , km/h), ambient temperature (x_3 , °C), wind speed (x_4 , km/h), waiting at shovel (x_5), waiting at dump (x_6), spotting (x_7), month (x_8), destination (x_9), and precipitation (x_{10}). They were selected because mining engineers observed these variables at mine sites and used them in previous studies to build prediction models [13, 22]. Using the

training dataset as an instance, a detailed description of all inputs and the output is presented in Table 1. In Table 1, the variables are divided into continuous variables (the first five) and categorical variables (the last six). Amid them, the inputs related to truck haulage were provided by the mine data management system; the weather-related inputs were collected from the local meteorological observatory [42].

Nevertheless, there are still some limitations to the input variables included in the current dataset. Some important variables may have an impact on truck cycle time that have not been considered yet such as load speed and variables related to elevation (e.g., gradient and the proportion of distance traveled on gradients versus flat surfaces). According to Pasch and Uludag [46], load speed has a direct impact on load time, which is a critical component of truck cycle time. Likewise, the elevation changes along the haul route can influence truck speed and the driver's vision [13, 55]. Depending on the accessibility of relevant data, these additional inputs could be incorporated into future studies to enhance the accuracy and robustness of truck cycle time prediction models.

Table 1 lists the statistical information of the available variables in this study, including minimum, mean, median, maximum, and first (Q_1) and third (Q_3) quartiles. Besides, the linear correlation (r) between each input and output was calculated by the extensively used Pearson correlation coefficient method [64]. As shown in Table 1, the correlation between some inputs and truck cycle time is low or even

zero, such as wind speed (0), waiting at dump (0.05), and spotting (0.08). The contribution of these variables in constructing models may be insignificant, but instead, they may increase the computational time and reduce the model predictability [34]. This provided the rationale for conducting feature selection (remove or retain these variables) in this study. Furthermore, the statistical distribution features of all inputs and the output are presented in Fig. 2. Figure 2a–e shows the distribution feature of each continuous variable using a histogram with specific columns. The horizontal direction denotes the variables; the vertical direction represents the density (proportion), indicating the fraction of each column divided by the total number of data points. Figure 2f–k presents the boxplots of six categorical variables.

2.3 Feature Selection Approaches

Feature selection approaches are broadly classified into wrapped, embedded, and filter methods [8, 44]. This study carried out a comparative study of three feature selection approaches to determine the essential optimal subset of input features. The basic ideas are briefly introduced below. The wrapped method, such as RFE, usually adopts specific machine learning algorithms (e.g., linear model or RF) to assess subsets of input variables and select the best subset that produces the highest performance. RFE works by recursively removing the least essential input variables from a specific model (e.g., RF) according to the change in model

Table 1 Descriptions and statistical information for the output (y) and inputs (x_m) in this study

Variable	Type	Description	Min	Q_1	Median	Mean	Q_3	Max	r
Cycle time (y , min)	Continuous	The time for each truck to finish a haulage cycle	14.67	23.62	27.01	27.82	31.00	78.30	1
Haul distance (x_1 , km)	Continuous	The distance between loading and dumping sites	1.02	3.41	4.71	4.47	5.45	11.12	0.37
Empty speed (x_2 , km/h)	Continuous	The truck speed from a dumping site to a loading site	6.40	31.30	36.9	37.00	42.70	59.90	−0.19
Ambient temperature (x_3 , °C)	Continuous	The hourly ambient temperature at mine sites	−38	−9.80	2.30	0.68	12.20	32.80	0.13
Wind speed (x_4 , km/h)	Continuous	The hourly wind speed (at 10 m) at mine sites	0	2	2	5.71	8.30	34	0
Waiting at shovel (x_5)	Categorical	Without and with wait time at a shovel for each truck cycle: 2 labels (0 and 1)	0	0	0	0.49	1	1	0.27
Waiting at dump (x_6)	Categorical	Without and with wait time at a dumping site for each truck cycle: 2 labels (0 and 1)	0	1	1	0.96	1	1	0.05
Spotting (x_7)	Categorical	Without and with spotting time for each truck cycle: 2 labels (0 and 1)	0	0	1	0.63	0.63	1	0.08
Month (x_8)	Categorical	12 months of the year: 12 labels (1–12)	1	4	7	6.55	10	12	0.15
Destination (x_9)	Categorical	Three dumping sites of truck haulage at mine sites: the labels (1, 2, and 3)	1	2	2	2.19	3	3	0.14
Precipitation (x_{10})	Categorical	Without and with hourly precipitation: 2 labels (0 and 1)	0	0	0	0.05	0	1	0.10

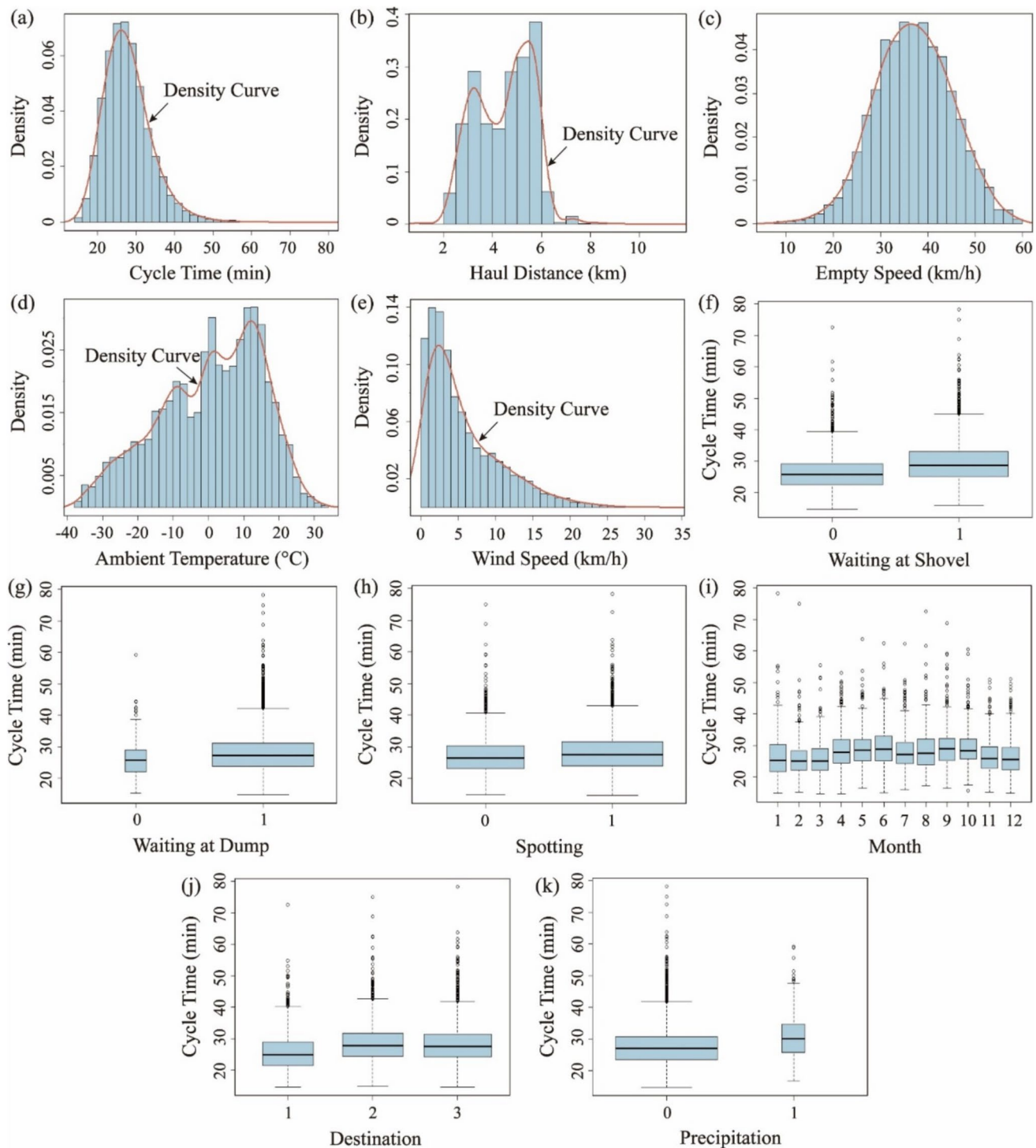


Fig. 2 Distributions with density curves and boxplots for the **a** output and **b–k** input variables

accuracy or some other criterion (e.g., regression coefficients) until the optimal number of inputs is achieved [6]. The embedded method directly incorporates feature selection into the model training of a machine learning algorithm, such as DT. DT selects input variables for the best division

at the root and internal nodes, simultaneously, it ranks the relative variable importance using some criterion (e.g., Gini index [65]). Unlike the wrapped and embedded methods, the filter method, such as ANOVA, uses statistical tests (e.g., significance tests) to estimate the relevance of input

variables, which is independent of any particular machine learning method. ANOVA computes the p -value when sequentially adding each input into a prediction model [51]. Meanwhile, the inputs with the lowest p -values (e.g., <0.01) are retained, indicating a strong correlation with the output.

2.4 Clustering Analysis Algorithms

Based on the density curves in Fig. 2, the truck haulage data show multi-peak distributions (e.g., haul distance and ambient temperature). This means that the data may have subgroups, which provides the rationale for selecting the clustering techniques, i.e., K-means and GMM. This information provides new insights for mining engineers to deal with data at other mine sites where truck haulage is the dominant means of transporting coal or other minerals. In this study, the K-means and GMM are two extensively applied unsupervised clustering techniques since they are easy to implement and efficient for dealing with massive data [11]. They both identify several classes from a data population and assign data points with more similarities to the same group [21]. The key distinction between the two techniques lies in the principles of assigning data points to classes. K-means assumes that each data point falls into the specific class where the centroid is closest to it. This centroid is updated iteratively until the squared distance sum between the centroid and each data point is minimized [36]. Unlike K-means, GMM is known as a probability-based clustering approach that assigns data points to a specific class when they have the maximum class posterior probability [27]. Detailed information on K-means and GMM is described below.

2.4.1 K-Means

K-means is a distance-based unsupervised clustering approach, which segments a data population into K subsets based on the difference in distance between data points. Assuming a group of data points $\{X_1, X_2, \dots, X_n\}$, where X_i indicates the i th data point. K-means finds the centroids $\{c_1, c_2, \dots, c_K\}$ for K classes that minimize the squared sum of the distance between a data point X_i and its closest centroid c_k . This distance $Dis(X_i, c_k)$ can be calculated as follows [33]:

$$Dis(X_i, c_k) = \sqrt{(X_i)^2 - (c_k)^2} \quad (1)$$

Initially, K-means randomly selected K centroids for the total data points. Next, the distance between the initial centroid and each data point is computed using Eq. (1). Under the principle of the nearest centroid, the data are assigned to the same latent class that is labeled as C_k , which can be expressed as [36]

$$C_k = \operatorname{argmin}_{k \in \{1, 2, \dots, K\}} Dis(X_i, c_k) \quad (2)$$

After that, K-means recalculate the locations of K centroids once all data points are assigned to the class C_k according to the following formula [33]:

$$c'_k = (\sum_{X \in C_k} X) / p \quad (3)$$

where p denotes the total number of data points in a specific class after K-means clustering.

K-means conducts this process iteratively (i.e., calculating distance and finding new centroids) until these centroids no longer change. Moreover, although the number of K ($K > 1$) is set by users according to corresponding clustering requirements, the optimal number can be determined by the gap statistic method, which is suggested by Yuan and Yang [61] and Sinaga and Yang [53].

2.4.2 Gaussian Mixture Modeling

GMM has shown its performance in many aspects of handling massive data, such as reservoir characterization [60], streamflow prediction [43], wind power forecast [25], and heat load pattern recognition [37]. GMM models the distribution of a data population by assuming that it comprises a mixture of Gaussian distributions. In other words, GMM assumes that the data population is generated by finite subsets (i.e., latent classes), each with its own Gaussian distribution [21]. The distribution function of the mixture of Gaussians can be formulated as [32]

$$P(y|x, \emptyset) = \sum_{k=1}^K \pi_k p_k(y|x, \mu_k, \Sigma_k) \quad (4)$$

where $P(y|x, \emptyset)$ indicates the distribution function of data points. $\emptyset$ denotes the parameter set $\{\pi_k, \mu_k, \Sigma_k\}$ of the distribution function. K is the number of latent classes. π_k is the mixture coefficient (non-negative) of the k th latent class with $\sum_{k=1}^K \pi_k = 1$. $p_k(y|x, \mu_k, \Sigma_k)$ is the distribution function of the k th class. μ_k and Σ_k are means and variances.

GMM aims to estimate the parameter set that constitutes the mixture of Gaussians. To achieve this aim, GMM uses a two-step strategy, which is known as expectation–maximization (EM). In the E-step, the probabilities (γ_{nk}) assigned to latent classes are calculated for each data point [32]:

$$\gamma_{nk} = \frac{\pi_k p_k(y_n|x_n, \mu_k, \Sigma_k)}{\sum_{k=1}^K \pi_k p_k(y_n|x_n, \mu_k, \Sigma_k)} \quad (5)$$

In the M-step, the parameter set is estimated with the γ_{nk} through maximizing the log-likelihood ($\log L$) [32]:

$$\log L = \sum_{n=1}^N \log(P(y|x, \phi)) = \sum_{n=1}^N \log\left(\sum_{k=1}^K \pi_k p_k(y|x, \mu_k, \Sigma_k)\right) \quad (6)$$

where $\pi_k = \frac{1}{N} p_k = \frac{1}{N} \sum_{n=1}^N \gamma_{nk}$. N refers to the number of data points. In GMM, the EM algorithm iteratively estimates the parameters until the maximized log L is attained. Finally, the optimal number of latent classes is obtained by the minimization of the Bayesian information criterion (BIC) [41]:

$$BIC = -2\log L + M\log N \quad (7)$$

where M indicates the number of parameters.

2.5 Machine Learning Algorithms and Performance Metrics

This study used three extreme machine learning algorithms (ELM, ERT, and XGBoost) to build prediction models of truck cycle time in open-pit mine haulage. These extreme algorithms are characterized by computational efficiency, whose principles are briefly introduced below. ELM is a neural network (NN) algorithm consisting of three layers: an input layer, an output layer, and a hidden layer [24], as depicted in Fig. 3. ELM can handle massive amounts of data, solve classification and regression tasks, and construct complex nonlinear relationships between inputs and outputs [29]. Unlike traditional NNs, the built-in parameters (e.g., weights between input and hidden nodes) in ELM are not required to be tuned because these

parameters are randomly initialized and remain constant throughout the training process [18]. This makes ELM extremely efficient and fast compared to other ANNs. ERT is a tree-based ensemble algorithm that integrates numerous DTs (see Fig. 4) for regression and classification tasks [48]. ERT is similar to RF, which builds a series of trees in a parallel way based on a bagging technique (i.e., bootstrapping and aggregation techniques) to improve model performance [22]. “Bootstrapping” indicates a sampling method that randomly samples subsets from the initial dataset with a replacement for training DTs. Meanwhile, a sample of input variables is arbitrarily selected at each DT node for best splitting. “Aggregation” is the averaging of the decisions across all DTs. Unlike RF, ERT makes splits at nodes based on entirely random input variables and thresholds. This results in more diverse and faster tree growth in ERT, thus enhancing model accuracy and generalization capability [62]. XGBoost is another tree-based ensemble method for solving supervised learning problems, such as regression and classification [14]. Unlike ERT, XGBoost builds a large number of DTs in a sequential manner based on a boosting technique, each of which attempts to learn and correct the errors of the preceding tree. In addition, XGBoost performs fast computation mainly because it adopts some built-in techniques, such as approximate greedy algorithm, parallel processing, and tree pruning [14]. These techniques help to prevent overfitting problems, improve model performance, and make the XGBoost algorithm more efficient [3].

Fig. 3 Typical architecture of the ELM model

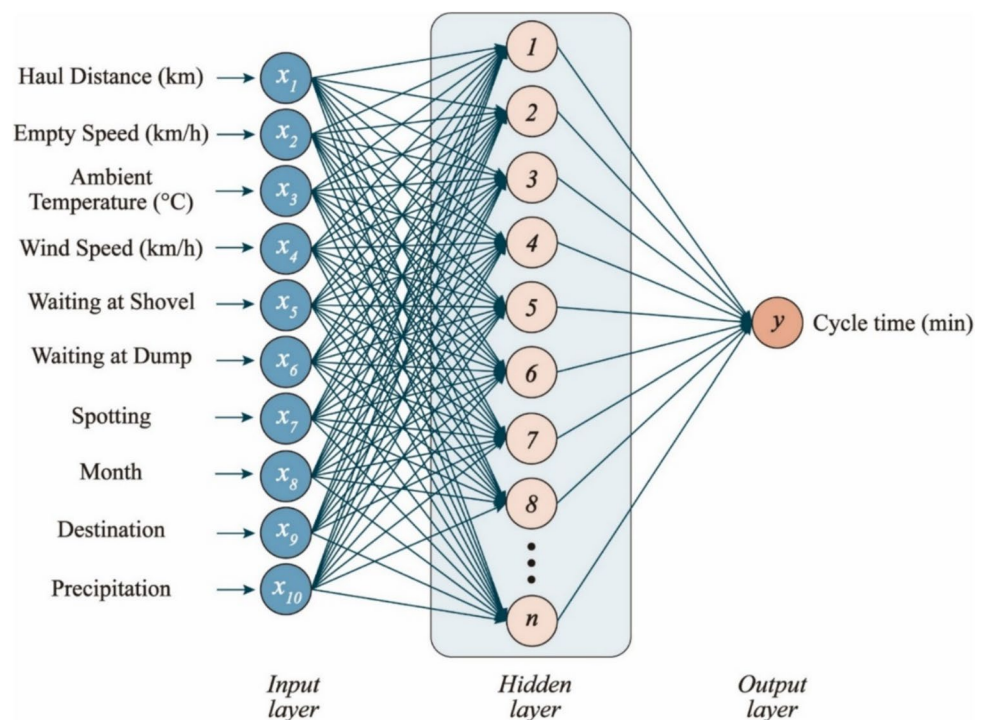
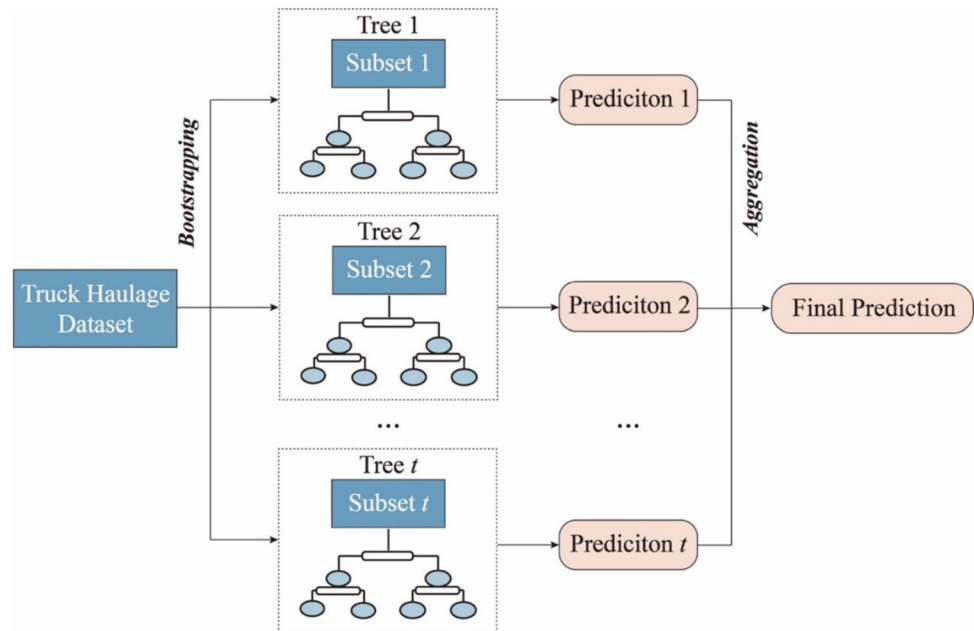


Fig. 4 Typical architecture of the ensemble tree-based models



These extreme models were constructed in RStudio with installed packages “elmNN,” “ranger,” “xgboost,” and “caret.” To evaluate the model performance, three quantitative indicators were used: MAE, RMSE, and R^2 , which are given below [30]:

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N |y_n - \hat{y}_n| \quad (8)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2} \quad (9)$$

$$R^2 = 1 - \frac{\sum_n (y_n - \hat{y}_n)^2}{\sum_n (y_n - \bar{y}_n)^2} \quad (10)$$

where y_n is the measured cycle time; $\hat{y}_n$ denotes the predicted cycle time, and $\bar{y}_n$ represents the mean of measured cycle time. MAE is the mean of absolute errors between the measured cycle time and the predicted cycle time. RMSE is the standard deviation of the errors between the measured and the predicted cycle time. R^2 is a measure of the goodness of fit, ranging from zero to one.

3 Results and Discussion

3.1 Feature Selection Analysis

Three feature selection approaches, including DT, ANOVA, and RFE, were availed in this study to remove redundant information (i.e., irrelevant input variables) from the

datasets. Then, the input variables chosen by each algorithm were applied to build and compare the ML model (e.g., XGBoost) to determine the best feature selection method [34]. Using the RFE algorithm as an example, Fig. 5 illustrates the results of feature selection under the principle of minimizing RMSE values. In Fig. 5, the number of input variables is increased from one to ten. When the number reaches five, RMSE drops significantly from 5.90 to 5.19. It descends further to 5.00 when the number attains seven and cannot decrease as the number of input variables grows. Therefore, RFE retained seven input variables. These inputs (in Table 2) were haul distance, empty speed, waiting at

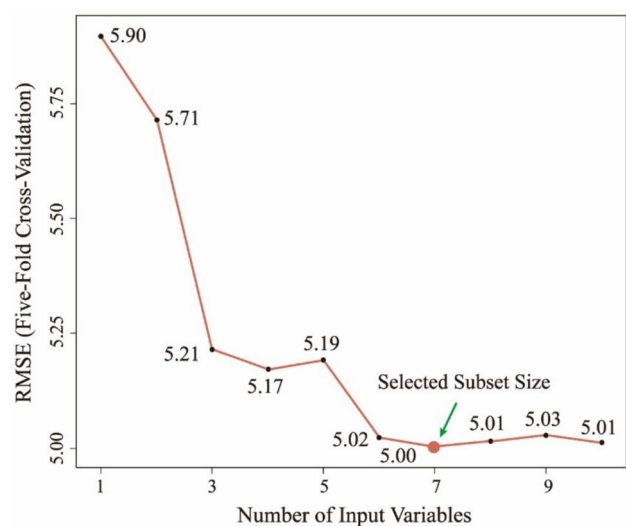


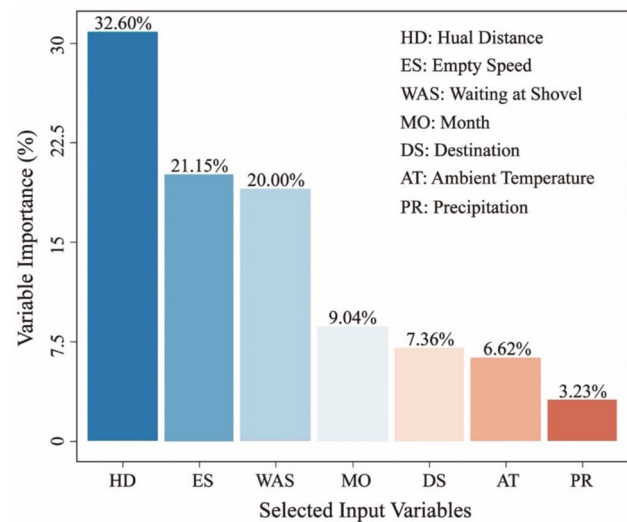
Fig. 5 Feature selection analysis using recursive feature elimination based on the training dataset

Table 2 Comparison of feature selection methods based on the XGBoost models

Method	Model	Subset size	Selected input	Performance metrics			Running time (s)
				RMSE	MAE	R^2 (%)	
Recursive feature elimination (RFE)	RFE-XGBoost	7	Haul distance, empty speed, waiting at shovel, month, destination, ambient temperature, precipitation	5.12	3.78	33.63	305.4
Analysis of variance (ANOVA)	ANOVA-XGBoost	7	Haul distance, empty speed, waiting at shovel, waiting at dump, spotting, precipitation, month	5.15	3.79	32.90	300.6
Decision tree (DT)	DT-XGBoost	6	Waiting at shovel, haul distance, empty speed, destination, ambient temperature, spotting	5.13	3.79	33.27	273.6
Null	XGBoost	10	Haul distance, empty speed, ambient temperature, wind speed, waiting at shovel, waiting at dump, spotting, month, destination, precipitation	5.13	3.79	33.37	320.4

shovel, month, destination, ambient temperature, and precipitation. From Table 2, ANOVA also selected seven input variables, but they were a new combination of haul distance, empty speed, waiting at shovel, waiting at dump, spotting, precipitation, and month. Compared to RFE and ANOVA, DT only kept six input variables, including waiting at shovel, haul distance, empty speed, destination, ambient temperature, and spotting. Afterward, these three subsets of inputs and the original inputs were involved in constructing four XGBoost models (i.e., RFE-XGBoost, ANOVA-XGBoost, DT-XGBoost, and XGBoost), whose performance was evaluated by the testing dataset. In Table 2, the RFE-XGBoost model has the smallest RMSE (5.12), MAE (3.78), and the highest R^2 (33.54%), which performs better than the DT-XGBoost (5.13, 3.79, and 33.27%), ANOVA-XGBoost (5.15, 3.79, and 32.90%), and XGBoost (5.13, 3.79, and 33.37%) models. Moreover, the running time of the RFE-XGBoost model (305.4 s) was lower than that of the XGBoost model (320.4 s), although it was longer than the ANOVA-XGBoost (300.6 s) and DT-XGBoost models (273.6 s). Akin to the study by Liu et al. [34], they adopted three feature selection methods (including the RFE algorithm) to eliminate irrelevant inputs and trained the XGBoost models for estimating the buildings' energy consumption. In terms of the model's performance and running time, the RFE-XGBoost had the highest R^2 (72.8%) and shortest running time (262 s). In short, RFE was the best feature selection method that improved the model performance and removed redundant variables, thus reducing the computational cost.

Furthermore, RFE determined the variable importance of the selected seven inputs by iteratively removing the least important features and evaluating the built-in model's performance [6]. As shown in Fig. 6, the importance scores (in

**Fig. 6** Variable importance analysis based on recursive feature elimination

percentage) of these inputs are ranked as follows: haul distance (32.60%) > empty speed (21.15%) > waiting at shovel (20.00%) > month (9.04%) > destination (7.36%) > ambient temperature (6.62%) > precipitation (3.23%). Among these inputs, haul distance, empty speed, and waiting at shovel were the most critical variables affecting truck cycle time. At mine sites, mining engineers usually built a fitted line (i.e., prediction model) between truck cycle time and haul distance because they observed that haul distance had a dominant impact on truck cycle time [12]. After haul distance, empty speed had the second-highest importance score (21.25%) since it can determine the travel time (a portion of cycle time) between dumping and loading sites [49]. Waiting

at shovel was a binary variable (zero and non-zero) in this study, with an importance of 20.00%. In accordance with Ercelebi and Bascetin [19], truck cycle time consisted of loading, hauling, dumping, returning time, and waiting time at dumps or shovels. Hence, the increase in waiting time contributes to a growth in cycle time. Moreover, other inputs influenced the truck cycle time to some extent. The variable importance of month was 9.04%, which may be attributed to the variations of time-related factors such as ambient temperature and precipitation in different months. For example, according to Ma et al. [38], truck tire temperature escalated from 54 to 69 °C when the ambient temperature at mine sites increased from 20 to 40 °C. This led to a reduction in tire performance (e.g., fatigue), thus increasing truck cycle time and decreasing truck productivity [39]. Due to the potential effect of multicollinearity [45], the importance of ambient temperature (6.62%) and precipitation (3.23%) was relatively lower than that of month (9.04%). Lastly, destination had an importance of 7.36%, which may be because destination is associated with truck haul length, thus affecting truck cycle time [9]. In addition, the importance can be assessed by investigating the effect of input size on the prediction performance, as shown in Table 3. For instance, XGBoost showed the largest improvement in R^2 (from 17.55 to 33.54%) when the model involved haul distance, indicating that it was the most influential input. This agreed with our previous studies [21, 24]. In brief, the RFE selected seven input variables and provided importance scores for each input. These scores help mining engineers gain insight into the main factors influencing truck cycle time at mine sites, leading to more accurate predictions of truck cycle time and the estimation of overall mine productivity [22].

3.2 Clustering Analysis Based on Truck Haulage Data

To improve truck cycle time predictions, this study used K-means and GMM to classify the massive truck haulage data (including the seven selected inputs by RFE). The

clustering analysis of these two techniques will be presented as follows.

3.2.1 K-Means Clustering Analysis

Figure 7 shows the clustering results using K-means based on the truck haulage data (training dataset). According to the gap statistic method [57], two classes were ultimately recommended as the optimal number of classes. In Fig. 7, these two classes are visualized in a two-dimensional scatter plot with two primary principal components (PC1 and PC2, with weights of 24.4% and 17.9%). This is because principal component analysis extracts predominant features from all input variables, which can be utilized to intuitively display the data distributions by a two-dimensional (2D) graphical demonstration [23, 36]. As presented in Fig. 7, it can be noted that Class 1 and Class 2 are almost entirely divided

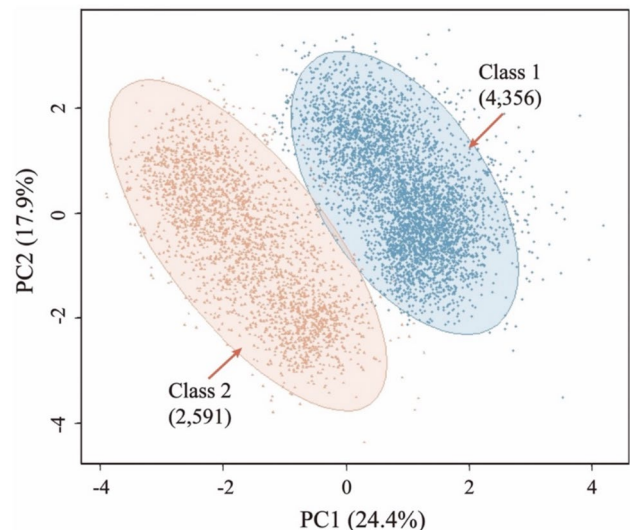


Fig. 7 Identifying two classes from the K-means analysis (distribution of data points in each class represented by a 2D scatter diagram with two major principal components)

Table 3 Different input sizes of input variables and the resulting model performance

XGBoost	Input size	Performance metrics		
		RMSE	MAE	R^2 (%)
Model 1	Precipitation	6.28	4.76	0.63
Model 2	Precipitation + ambient temperature	6.17	4.69	3.37
Model 3	Precipitation + ambient temperature + destination	6.11	4.61	5.21
Model 4	Precipitation + ambient temperature + destination + month	6.10	4.59	5.68
Model 5	Precipitation + ambient temperature + destination + month + waiting at shovel	5.88	4.43	12.41
Model 6	Precipitation + ambient temperature + destination + month + waiting at shovel + empty speed	5.70	4.28	17.55
Model 7	Precipitation + ambient temperature + destination + month + waiting at shovel + empty speed + haul distance	5.12	3.78	33.54

into two datasets with different centroids. In Class 1, the number of data points was 4356, whereas the number in Class 2 was 2591. Moreover, using eight boxplots, Fig. 8 illustrates the statistical characteristics of inputs and output in these two classes. The first four represent the relationships among the classes and continuous variables; the last four show the relationships among the classes and categorical variables. As can be seen from these boxplots, for the continuous variables, only ambient temperature (Fig. 8d) is well split into two classes, with mean values of 10.03 °C and −15.04 °C, respectively. All other continuous variables have overlapping distributions in two classes. Likewise, for the categorical variables, a clear relationship can be uniquely observed between months and classes (Fig. 8f). The data points in Class 1 are almost distributed from March to October, while the data points in Class 2 are mainly present in January, February, November, and December. This corresponds to the values of ambient temperature in two classes, representing the warm and cold temperatures in Northern Alberta, Canada. This can also be observed in Fig. 9, which displays the scattering relationships for the continuous variables and cycle time. Both in Classes 1 and 2, a positive correlation was achieved between haul distance and cycle time, while a negative correlation was established between empty speed and cycle time. This is akin to our previous studies [20, 58]. For ambient temperature, in Fig. 9c, there is a clear boundary (~ -2 °C) between Classes 1 and 2. Above -2 °C is Class 1; the other side is Class 2. Therefore, it can be concluded that these two classes recognized by K-means from the massive truck haulage data were closely related to the ambient temperature. Similar results were discovered by Liu et al. [36], who applied K-means to classify the single-crystal superalloy creeping data and developed prediction models of creep rupture life. The research showed that K-means identified eight homogeneous classes that were intimately associated with the creep mechanisms, which further improved the model accuracy of creep rupture life with an increase in R^2 from 71.02 to 91.76%.

3.2.2 GMM Clustering Analysis

In addition to K-means, this study adopted the GMM algorithm to classify massive truck haulage data. Following the principle of minimizing the BIC value [40], the optimal amount of classes was ascertained. As a result, three classes were extracted from the training dataset by GMM, as shown in Fig. 10. These three classes were visualized in a two-dimensional scattering diagram between cycle time and each continuous input variable (i.e., haul distance, empty speed, and ambient temperature). Taking Fig. 10a as an example, the data points in each class were covered by an ellipse that represented a Gaussian distribution [52]. The number of data points varied in these three classes. Classes 1 (3471) and 2

(3099) had more than 3000 data points, while Class 3 contained 377 data points. Despite this, truck cycle times grow accordingly in each class as haul distance increases. This is in agreement with the site observations [13]. Similarly, Fig. 10b and c presents the three classes and the relationship between cycle time and other two continuous input variables. Furthermore, Fig. 11 shows the data distribution in each class using histograms and boxplots. After conducting GMM analysis, a large amount of truck haulage data came from three different populations, each of which followed a standard Gaussian distribution (e.g., as evidenced by the density curves) according to the definition of GMM [21]. Figure 11c presents the relationship between cycle time and classes; Fig. 11d–g displays the relationship between each categorical variable (including waiting at shovel, month, destination, and precipitation) and classes. In Fig. 11d–g, it can be seen that the three classes have no significant correlation with the categorical variables; despite this, truck cycle time significantly varied among these classes in Fig. 11c, showing the order: Class 3 > Class 2 > Class 1 (e.g., the mean values were 41.26 min, 30.44 min, and 24.02 min). This indicates short, medium, and long truck cycle times at mine sites. This means that these classes extracted by GMM were well linked to truck cycle time and may relate highly to the continuous variables. This is comparable with the studies of Ni et al. [43] and Lu et al. [37]. In the study of Ni et al. [43], they adopted GMM to extract two latent classes (low and high streamflow) from massive hydrological data and built an XGBoost model for forecasting monthly streamflow. The results presented that the model's performance increased by about 11% based on GMM analysis. Similarly, Lu et al. [37] classified building heating data using GMM to identify sub-patterns (including six classes). After that, they trained models for predicting the hourly heating load, whose performance was enhanced by approximately 20% because of GMM analysis.

3.3 Evaluation of Feature-Optimized Machine Learning Models

In Sect. 3.1, seven input variables were selected by RFE. Based on these inputs and the output, in Sect. 3.2, two unsupervised clustering techniques, including K-means (abbreviated as KM only in models) and GMM, were adopted to identify classes from the massive truck haulage data. The labels of these classes constituted additional variables and were considered as new inputs (i.e., new categorical variables) incorporated with the selected inputs to become new training datasets [21, 22]. Therefore, the training data for extreme machine learning models included eight inputs based on feature optimization (i.e., feature selection and unsupervised clustering). After that, the ELM, ERT, and XGBoost algorithms were applied to establish prediction

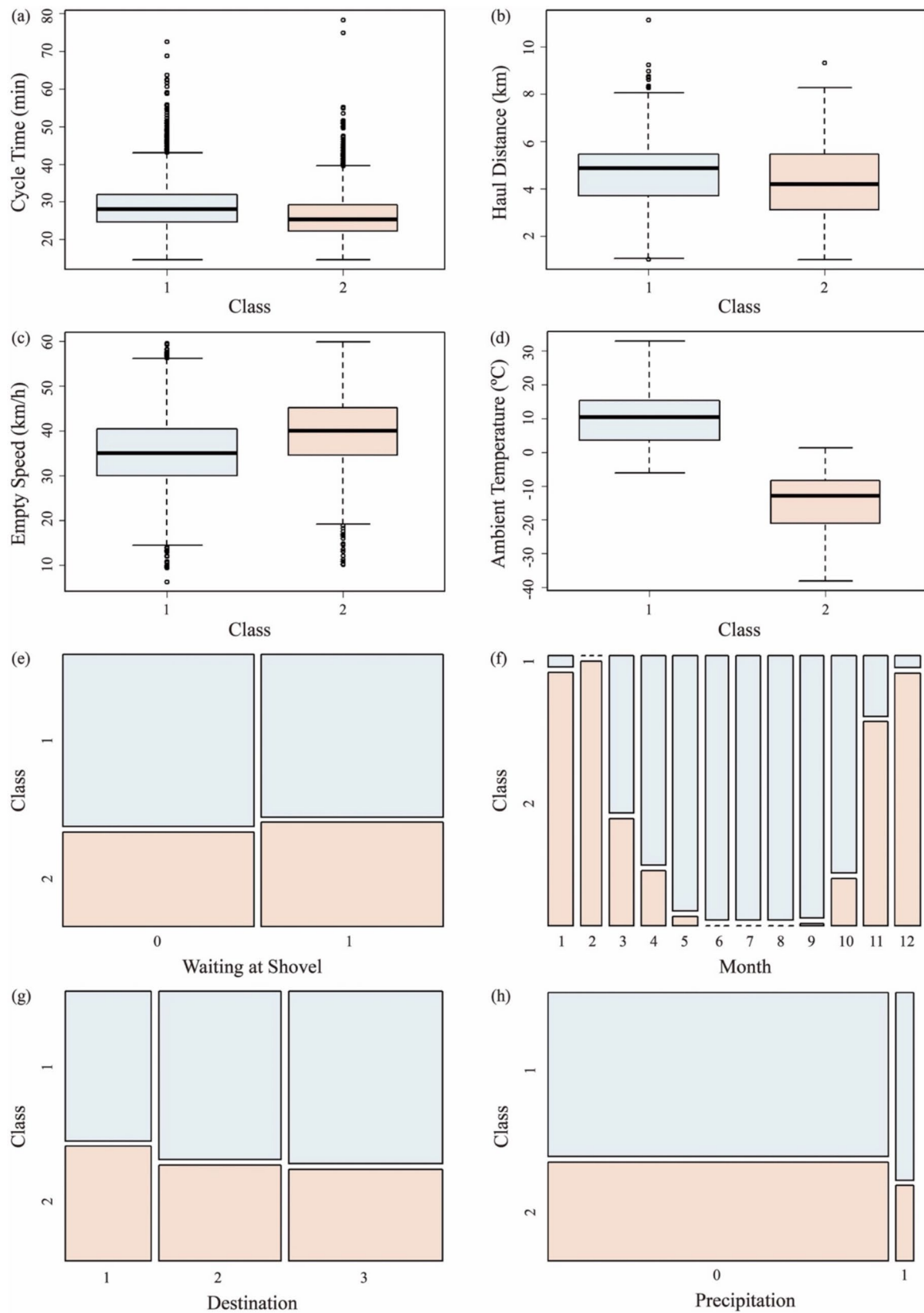


Fig. 8 Boxplots of each input variable to corresponding classes based on the K-means clustering analysis

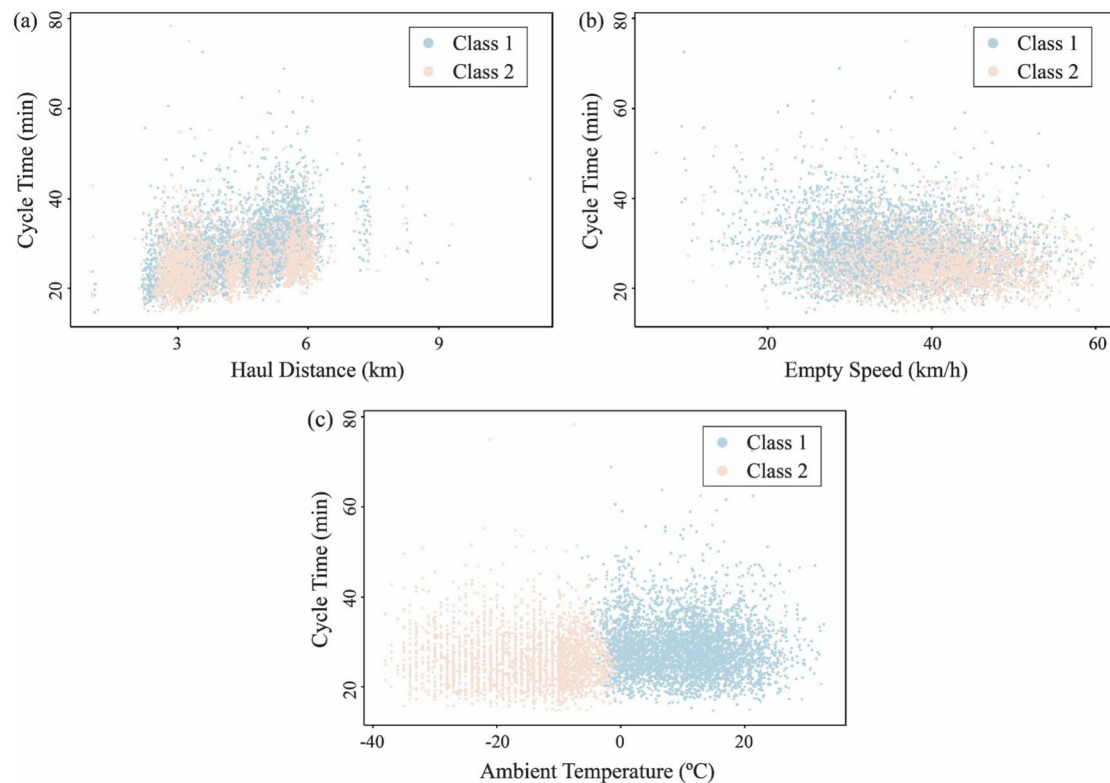


Fig. 9 Scatterplots of the continuous variables (haul distance, empty speed, and ambient temperature) and cycle time in two classes based on the K-means clustering analysis

models of truck cycle time. With the new training datasets, six hybrid prediction models were constructed, which were referred to as the KM-RFE-ELM, KM-RFE-ERT, KM-RFE-XGBoost, GMM-RFE-ELM, GMM-RFE-ERT, and GMM-RFE-XGBoost models. Moreover, three baseline models (RFE-ELM, RFE-ERT, and RFE-XGBoost) were built on the original dataset (i.e., the seven inputs and the output) and compared to the clustering analysis-based models.

Figure 12 exhibits the scatter points of measured cycle time (horizontal) from the testing datasets and the predicted cycle time (vertical) from these nine prediction models. The more evenly distributed these scatter points are along the 45° diagonal, the better the forecasting is [22]. In Fig. 12a–f, the scatter points from these six prediction models (baseline and K-means-based models) are dispersedly distributed along the diagonal, especially for the longer truck cycle time. This is because many complex factors contribute to the long cycle time at mine sites, such as weather changes, equipment overhaul, work shifts, and road maintenance [1, 24], however, these factors were not included yet as inputs due to the data availability, resulting in poor predictions of long cycle time. Compared to Fig. 12a–f, Fig. 12g–i shows the scatter points that are uniformly distributed along the diagonal, indicating that the GMM-based models worked well in predicting cycle time. It can also be seen in Fig. 13 through three

performance metrics: RMSE, MAE, and R^2 . For instance, GMM-RFE-XGBoost had the highest R^2 (80.37%) and the least RMSE (2.78) and MAE (1.98). Its performance was considerably higher than the KM-RFE-XGBoost (5.14, 3.80, and 33.06%) and RFE-XGBoost (5.12, 3.78, and 33.63%) models. Therefore, two findings can be drawn from this study: (1) GMM significantly improved the models' accuracy. This is mainly because the truck haulage data usually have a mixture of Gaussian distributions, referred to as multi-peaked Gaussian distributions (as shown in Fig. 2b and d) [7]. Depending on these specific peaks, GMM can recognize classes from massive data and extract corresponding latent variables [25]. In Sect. 3.2.2, the additional variable from GMM analysis was strongly connected with truck cycle time. As a consequence, the model performance can be enhanced when combining these latent variables as new inputs in building prediction models. This is consistent with our previous studies [21, 22], for example, the R^2 of the truck productivity prediction model was increased from 44 to 87% based on GMM analysis. (2) K-means was unable to increase the models' predictability. In Sect. 3.2.1, a categorical variable with two labels was extracted through K-means analysis, which showed a close relationship with ambient temperature. Nevertheless, ambient temperature has been considered an input variable in predicting truck cycle

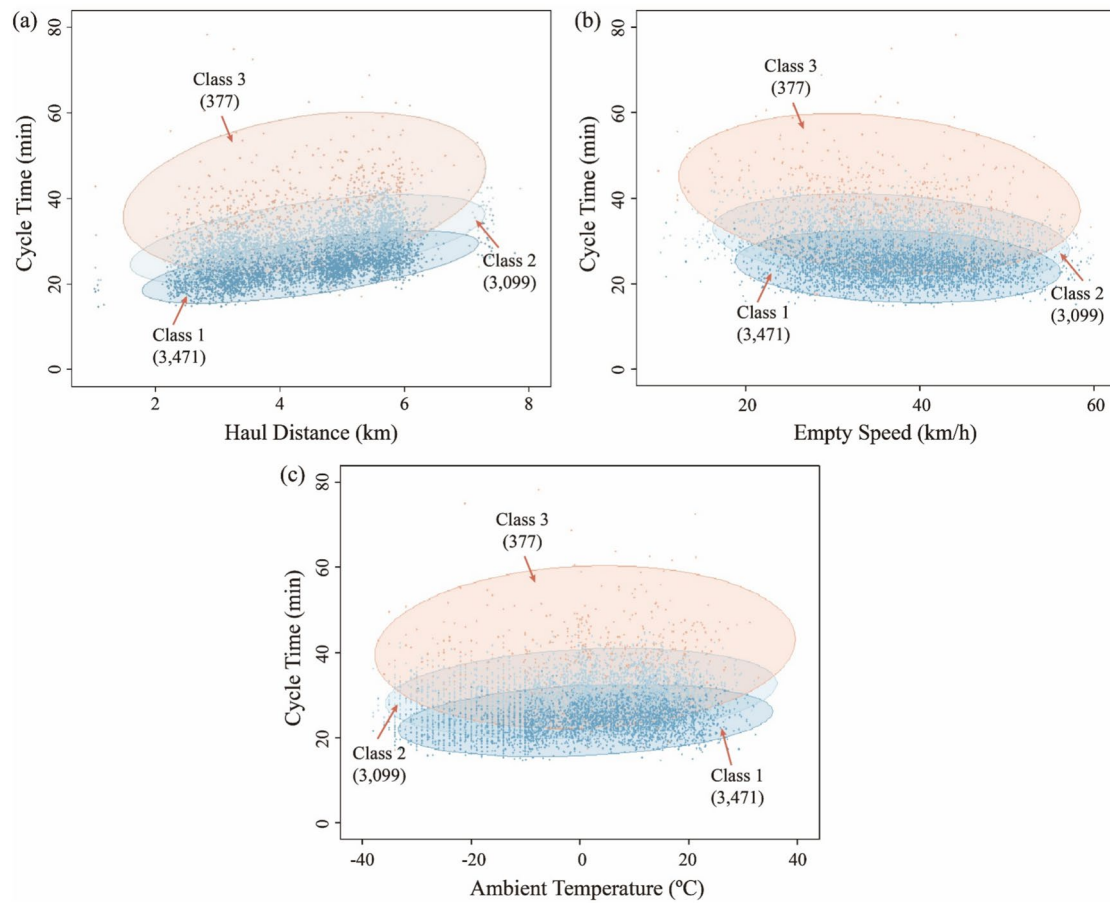


Fig. 10 Scatterplots of the continuous variables (haul distance, empty speed, and ambient temperature) and cycle time in three classes based on the GMM analysis (note: the ellipses that represented a Gaussian distribution)

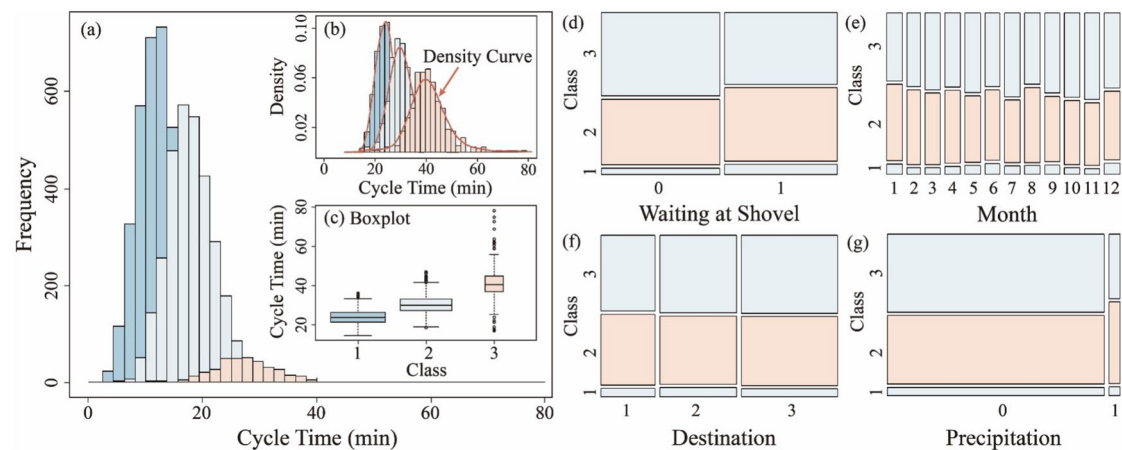


Fig. 11 Identifying three classes from the GMM analysis. **a** The data in each class follow standard Gaussian distributions, **b** density curves for each class, **c** a boxplot for three classes in cycle time, and **d–g** boxplots for three classes in the categorical input variables

time, whose variable importance was 6.62% (see Sect. 3.1). The inclusion of the categorical variable obtained from K-means analysis may increase the model's multicollinearity

and computational complexity, thus reducing the model's accuracy [45]. In summary, GMM performed better than K-means in enhancing the model predictability. This study

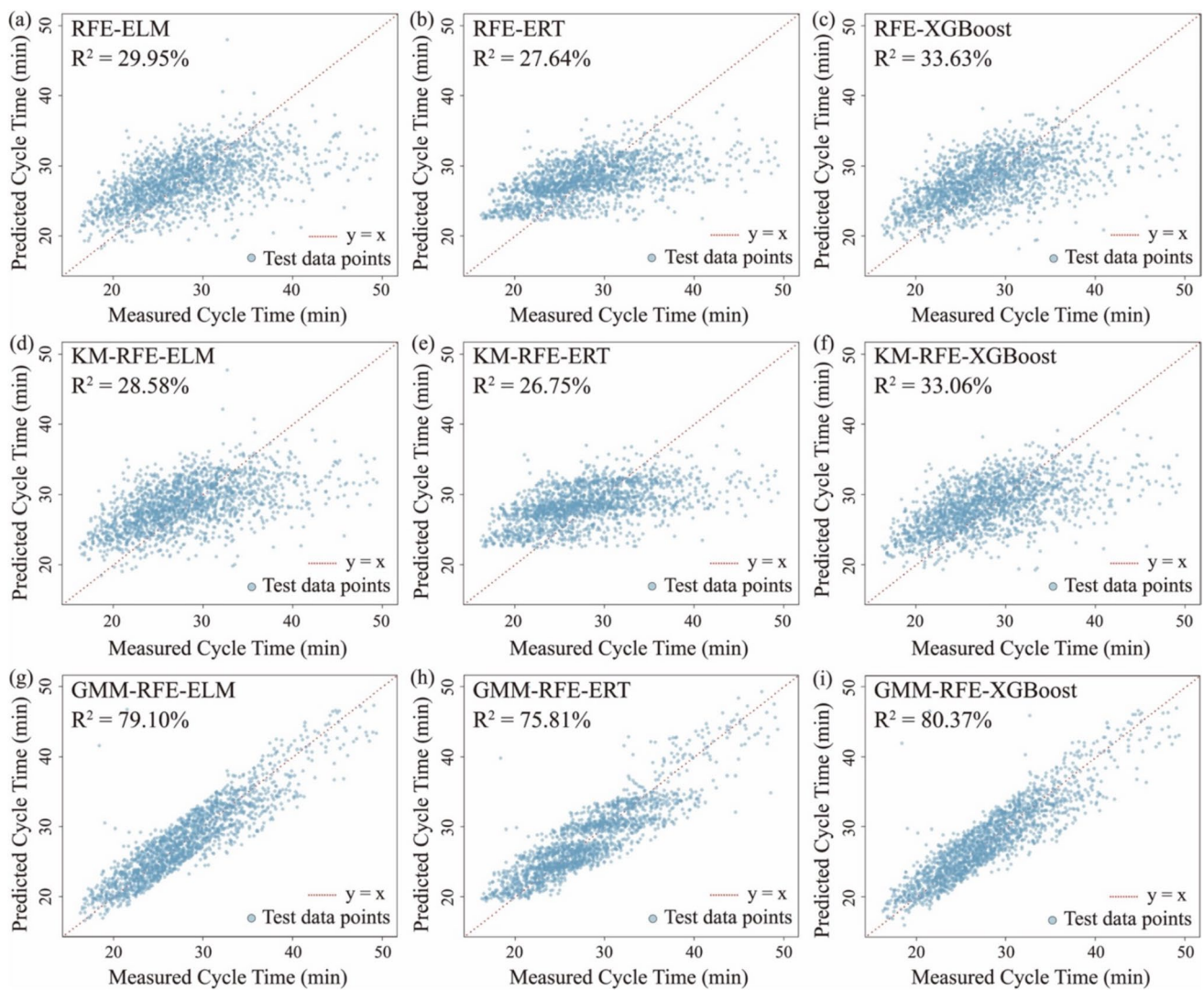


Fig. 12 Scatterplots of the predicted cycle time and measured cycle time. **a–c** The benchmark models (RFE-ELM, RFE-ERT, and RFE-XGBoost) without clustering analysis. **d–f** The RFE-ELM, RFE-

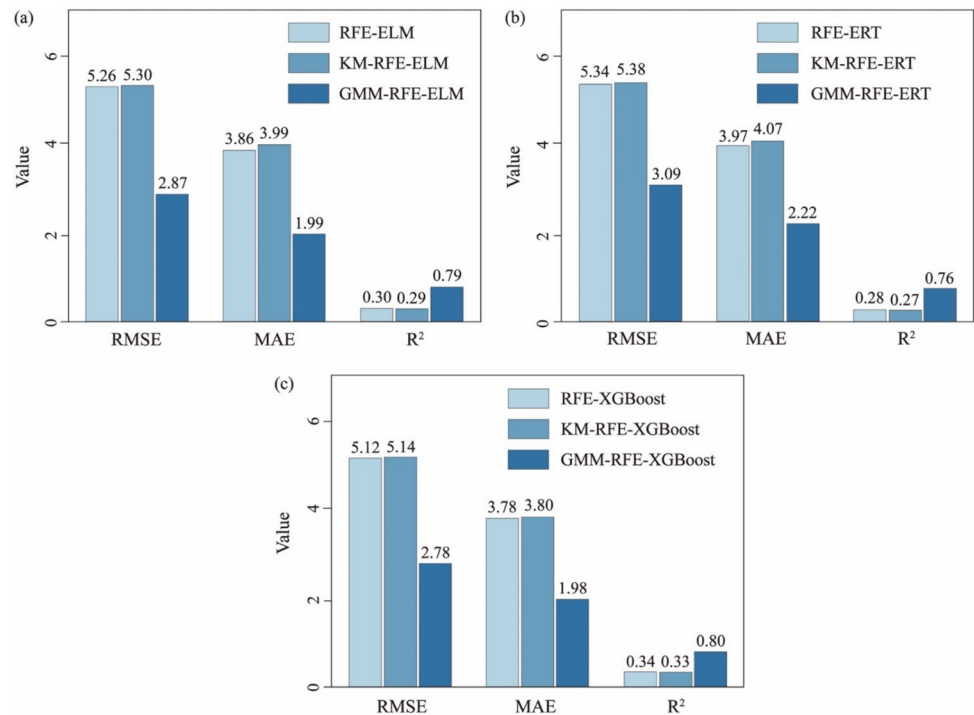
ERT, and RFE-XGBoost based on K-means (KM) analysis. **g–i** The RFE-ELM, RFE-ERT, and RFE-XGBoost based on GMM analysis

was similar to the results from Virupakshappa and Oruklu [59], they built prediction models for detecting and positioning flaw echoes in ultrasonic data based on three unsupervised clustering techniques, including K-means and GMM. The results showed that the detection accuracy achieved 93% with GMM, which was higher than that of K-means (89%).

Furthermore, an additional finding can be obtained by comparing these feature-optimized extreme machine learning models: the XGBoost algorithm outperformed the ELM and ERT algorithms in predicting truck cycle time. For example, for the baseline models, the R^2 of the RFE-XGBoost model was 33.63%, while the RFE-ELM and RFE-ERT models were 29.95% and 27.64%. This applies to the clustering-based models. For example, the R^2 of GMM-RFE-XGBoost was 80.37%, which was higher than

GMM-RFE-ELM (79.10%) and GMM-RFE-ERT (75.81%). This may be because XGBoost constructs a loss function and adopts a gradient descent method to find the optimal weights for each feature, thus increasing the model predictability [14]. In ELM, the weights are randomly chosen and optimized by a linear algorithm to reduce the residuals between predicted and actual values [29]. As for ERT, it simply averages the predictions over all randomized trees [26]. This is close to Cao et al. [10], who built the XGBoost and ELM models to predict the deformation and damage of super-high arch dams. The research proved that the XGBoost model's RMSE (0.90) was lower than the ELM model (1.10). Janizadeh et al. [31] also demonstrated that the XGBoost model ($R^2=92\%$) performed better than the ERT model ($R^2=91\%$) when mapping the flood hazard susceptibility. In short, the

Fig. 13 Comparisons of extreme machine learning models with and without clustering analysis. **a** Trained models using extreme learning machine (ELM). **b** Trained models using an extremely randomized tree (ERT). **c** Trained models using extreme gradient boosting (XGBoost)



GMM-RFE-XGBoost model exhibited the highest accuracy by comparing and evaluating the prediction models, which provides mining engineers with a more accurate method to estimate mining truck cycle time.

3.4 Rapid Estimation of Feature-Optimized Machine Learning Models

In the last section (Sect. 3.3), the GMM-based models (i.e., GMM-RFE-XGBoost, GMM-RFE-ELM, and GMM-RFE-ERT) performed best in estimating truck cycle time at mine sites. In addition to that, these extreme machine learning models had superiority with respect to computational efficiency. Figure 14 shows the changes in running time and performance of these three models when tuning one built-in hyperparameter. Using Fig. 14a as an example, the vertical axis includes the RMSE values and running time; the horizontal axis is the number of hidden neurons (nhid). The green dashed line indicates the location of the lowest RMSE and the corresponding optimal hyperparameter. As shown in Fig. 14a, the RMSE value drops as the number of hidden neurons increases. When the RMSE reached the minimum value (2.87), the optimal number of hidden neurons was 22; meanwhile, the running time was only 0.36 s. Likewise, for the GMM-RFE-ERT and GMM-RFE-XGBoost models, the optimal hyperparameters are displayed in Fig. 14b and c. The running times to build these two models were less than 1 s (0.61 s and 0.89 s) when tuning one built-in hyperparameter. Moreover, the ELM, ERT, and XGBoost algorithms usually contain

multiple hyperparameters that need to be tuned. Consequently, the total running time for each GMM-based model is listed in Table 4. For these three prediction models, the running times were 3.2 s, 183.0 s, and 314.4 s, respectively, in the order of GMM-RFE-ELM < GMM-RFE-ERT < GMM-RFE-XGBoost. Compared to these extreme models, the other commonly used models often need more computational cost. For example, the ANN models, such as the Bayesian regularized neural network (BRNN) and BPNN, took a long time (14,004.0 s and 9072.0 s) in our previous study [24]. Therefore, the extreme machine learning models were more computationally efficient for the rapid estimation of truck cycle time. The main reason for the fast assessment is the random initialization of weights between the layers in ELM [29], random selections of features for each tree and splitting thresholds in ERT [62], and parallel processing and approximate greedy algorithm for the best split in XGBoost [47]. The same results were found in the research by Sekhar Roy et al. [50], Chencho et al. [15], and Liu et al. [35]. They all demonstrated that EML, ERT, and XGBoost provided rapid estimations. For instance, Chencho et al. [15] constructed an ERT model for single-element damage quantification of civil engineering structures. The ERT model showed powerful performance in computation (the training time was 13.42 s) and prediction (e.g., $R^2 = 99.9\%$). To sum up, this study proposed three computationally efficient models that will facilitate mining engineers to rapidly estimate truck cycle time and make better decisions.

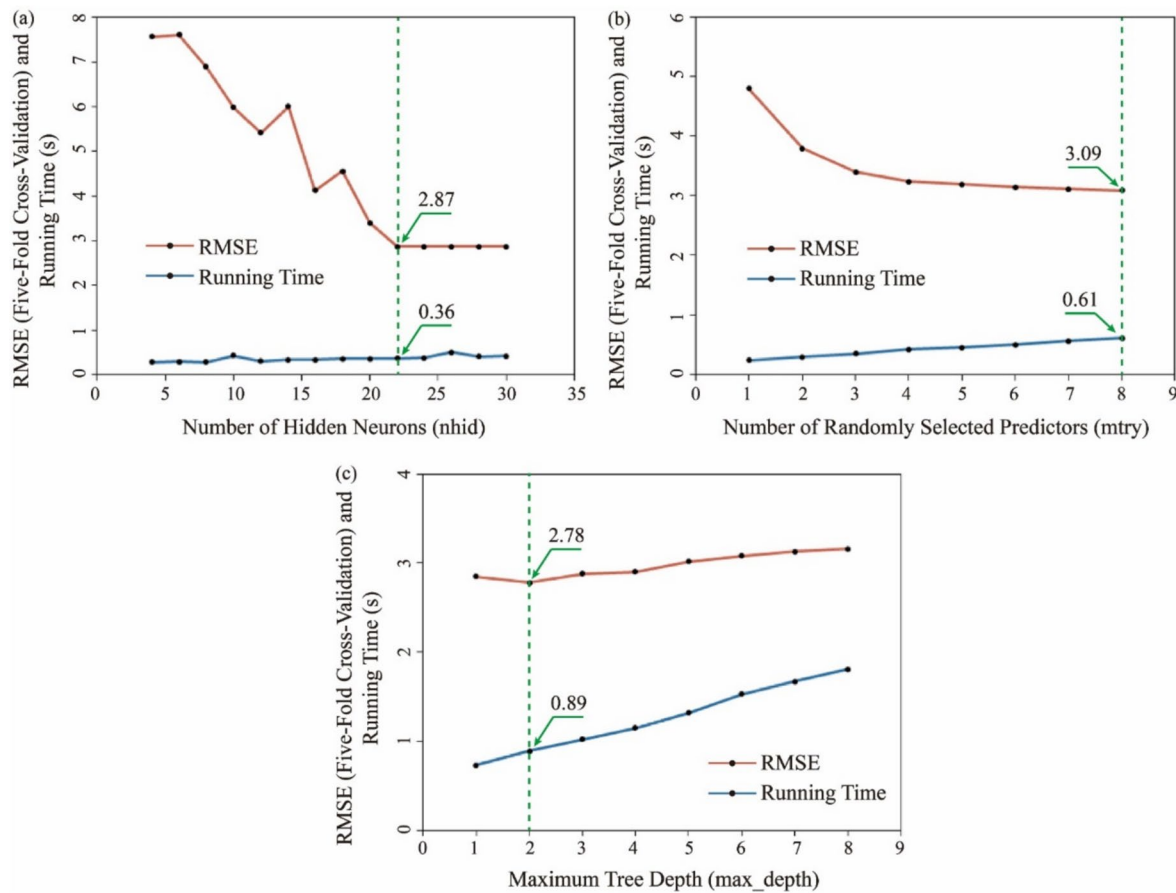


Fig. 14 Relationships of model performance (RMSE), running time of **a** GMM-RFE-ELM, **b** GMM-RFE-ERT, and **c** GMM-RFE-XGBoost models

Table 4 Running times of machine learning models between this and previous studies

Algorithm	Model	Input variable	Running time (s)	Reference
Bayesian regularized neural network (BRNN)	BRNN	Haul distance, empty speed, ambient temperature, waiting at shovel	14,004.0	Fan et al. [24]
Backpropagation neural network (BPNN)	BPNN		9072.0	
Random forest (RF)	RF		2726.4	
Gradient boosting regression (GBR)	GBR		1998.6	
Extreme learning machine (ELM)	GMM-RFE-ELM	Haul distance, empty speed, waiting at shovel, month, destination, ambient temperature, precipitation	3.2	This study
Extremely randomized tree (ERT)	GMM-RFE-ERT		183.0	
Extreme gradient boosting (XGBoost)	GMM-RFE-XGBoost		314.4	

4 Conclusion

Machine learning is known for handling massive data and providing fast computations, such as XGBoost, ELM, and ERT. For the first time, this study applied these machine learning algorithms to establish prediction models of truck cycle time based on massive and multidimensional data

in open-pit mine haulage. The proposed models leveraged advanced feature selection (e.g., RFE) and clustering (e.g., GMM) techniques to reduce computational costs and improve prediction accuracy. By identifying critical variables affecting truck cycle time, the models provide a foundation for real-time decision-making at mine sites. While this study focused on the prediction of truck cycle time, the results and methodology can serve as a basis

for future applications aimed at enhancing productivity (e.g., through optimizing haulage scheduling), optimizing costs (e.g., via cutting fuel consumption and maintenance), and improving safety (e.g., by mitigating congestion and operational risks). The main findings are concluded below:

- (1) RFE (the wrapped method) was the best method to select the optimal subset of input features, which improved the model performance and removed redundant variables. For example, the RFE-XGBoost model selected seven input variables from the original dataset, which achieved the lowest RMSE (5.12), MAE (3.78), and the highest R^2 (33.63%) compared to the ANOVA-XGBoost (5.15, 3.79, and 32.90%), DT-XGBoost (5.13, 3.79, 33.27%), and XGBoost (5.13, 3.79, and 33.37%) models.
- (2) Among the selected inputs, haul distance, empty speed, and waiting at shovel were the most critical variables affecting truck cycle time. The variable importance ranking presents here: haul distance (32.60%) > empty speed (21.15%) > waiting at shovel (20.00%) > month (9.04%) > destination (7.36%) > ambient temperature (6.62%) > precipitation (3.23%). These scores help engineers gain insight into the main factors influencing truck cycle time at mine sites, leading to more accurate predictions of truck cycle time and the estimation of overall mine productivity.
- (3) GMM performed better than K-means and significantly improved the models' accuracy. For instance, the GMM-RFE-XGBoost model had the highest R^2 , with values of 80.37%. Its performance was considerably better than the RFE-XGBoost (33.63%) and KM-RFE-XGBoost (33.06%) models.
- (4) The XGBoost algorithm outperformed the ELM and ERT algorithms in predicting truck cycle time. For example, the R^2 of the GMM-RFE-XGBoost model was 80.37%, which was higher than that of the GMM-RFE-ELM (79.10%) and GMM-RFE-ERT (75.81%).
- (5) The feature-optimized hybrid models provided rapid estimations of truck cycle time. For the GMM-based models, the runtimes were 3.2 s, 183.0 s, and 314.4 s for GMM-RFE-ELM, GMM-RFE-ERT, and GMM-RFE-XGBoost, respectively. Among them, the GMM-RFE-ELM had the shortest running time without decreasing significant accuracy. This will facilitate engineers to estimate truck cycle time and make decisions rapidly at open-pit mine sites.

Author Contribution Chengkai Fan: conceptualization, methodology, coding, writing—original draft, writing—review and editing,

visualization. Na Zhang: conceptualization, methodology, coding. Bei Jiang: conceptualization, methodology, resources, supervision, writing—review and editing. Wei Victor Liu: conceptualization, methodology, resources, supervision, writing—review and editing, funding acquisition.

Funding This work is supported by the Collaborative Research Project (RES0043251) and the Pilot Seed Grant (RES0049944) at the University of Alberta.

Data Availability The data that support the findings of this study are available from the corresponding author, WVL, upon reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

References

1. Alarie S, Gamache M (2002) Overview of solution strategies used in truck dispatching systems for open pit mines. *Int J Surf Min Reclam Environ* 16:59–76
2. Arachchilage CB, Fan C, Zhao J, Huang G, Liu WV (2023) A machine learning model to predict unconfined compressive strength of alkali-activated slag-based cemented paste backfill. *J Rock Mech Geotech Eng* 15:2803–2815
3. Arachchilage CB, Huang G, Fan C, Liu WV (2023) Forecasting unconfined compressive strength of calcium sulfoaluminate cement mixtures using ensemble machine learning techniques integrated with shapely-additive explanations. *Constr Build Mater* 409:134083
4. Arachchilage CB, Huang G, Zhao J, Fan C, Liu WV (2024) Hybrid extreme gradient boosting regressor models for the multi-objective mixture design optimization of cementitious mixtures incorporating mine tailings as fine aggregates. *Cement Concr Compos* 154:105787
5. Arciszewski TJ, Hazewinkel RRO, Dubé MG (2022) A critical review of the ecological status of lakes and rivers from Canada's oil sands region. *Integr Environ Assess Manag* 18:361–387
6. Bahl A, Hellack B, Balas M, Dinischiotu A, Wiemann M, Brinkmann J, Luch A, Renard BY, Haase A (2019) Recursive feature elimination in random forest classification supports nanomaterial grouping. *NanoImpact* 15:100179
7. Bishop CM (2006) Pattern recognition and machine learning. Springer, Verlag New York
8. Boratto THA, Cury AA, Goliatt L (2023) Machine learning-based classification of bronze alloy cymbals from microphone captured data enhanced with feature selection approaches. *Expert Syst Appl* 215:119378
9. Both C, Dimitrakopoulos R (2020) Joint stochastic short-term production scheduling and fleet management optimization for mining complexes. *Optim Eng* 21:1717–1743
10. Cao E, Bao T, Li H, Xie X, Yuan R, Hu S, Wang W (2022) A hybrid feature selection-multidimensional LSTM framework for deformation prediction of super high arch dams. *KSCE J Civ Eng* 26:4603–4616
11. Capó M, Pérez A, Lozano JA (2017) An efficient approximation to the K-means clustering for massive data. *Knowl-Based Syst* 117:56–69
12. Cervantes EG, Upadhyay SP, Askari-Nasab H (2019) Improvements to production planning in oil sands mining through

- analysis and simulation of truck cycle times, Mining Optimization Laboratory (MOL). University of Alberta, pp. 142–156
13. Chanda EK, Gardiner S (2010) A comparative study of truck cycle time prediction methods in open-pit mining. *Eng Constr Archit Manag* 17:446–460
 14. Chen T, Guestrin C (2016) XGBoost: a scalable tree boosting system, Proceedings of the 22Nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM Digital Library, New York, NY, USA, pp. 785–794
 15. Chencho, Li J, Hao H, Wang R, Li L (2022) Structural damage quantification using ensemble-based extremely randomised trees and impulse response functions. *Structural Control Health Monit* 29:e3033
 16. Daware S, Chandel S, Rai B (2022) A machine learning framework for urban mining: a case study on recovery of copper from printed circuit boards. *Miner Eng* 180:107479
 17. Demir S, Sahin EK (2023) An investigation of feature selection methods for soil liquefaction prediction based on tree-based ensemble algorithms using AdaBoost, gradient boosting, and XGBoost. *Neural Comput Appl* 35:3173–3190
 18. Dhini A, Surjandari I, Kusumoputro B, Kusiak A (2022) Extreme learning machine-radial basis function (ELM-RBF) networks for diagnosing faults in a steam turbine. *J Ind Prod Eng* 39:572–580
 19. Ercelesi SG, Bascetin A (2009) Optimization of shovel-truck system for surface mining. *J South Afr Inst Min Metall* 109:433–439
 20. Fan C, Arachchilage CB, Zhang N, Jiang B, Liu WV (2024) Machine learning with SHapley additive exPlanations for evaluating mine truck productivity under real-site weather conditions at varying temporal resolutions. *Int J Min Reclam Environ* 38:810–832
 21. Fan C, Zhang N, Jiang B, Liu WV (2022) Preprocessing large datasets using Gaussian mixture modelling to improve prediction accuracy of truck productivity at mine sites. *Arch Min Sci* 67:661–680
 22. Fan C, Zhang N, Jiang B, Liu WV (2023) Prediction of truck productivity at mine sites using tree-based ensemble models combined with Gaussian mixture modelling. *Int J Min Reclam Environ* 37:66–86
 23. Fan C, Zhang N, Jiang B, Liu WV (2023) Using deep neural networks coupled with principal component analysis for ore production forecasting at open-pit mines. *J Rock Mech Geotech Eng* 16:727–740
 24. Fan C, Zhang N, Jiang B, Liu WV (2023) Weighted ensembles of artificial neural networks based on Gaussian mixture modeling for truck productivity prediction at open-pit mines. *Min Metall Explor* 40:583–598
 25. Ge F, Ju Y, Qi Z, Lin Y (2018) Parameter estimation of a Gaussian mixture model for wind power forecast error by Riemann L-BFGS optimization. *IEEE Access* 6:38892–38899
 26. Geurts P, Ernst D, Wehenkel L (2006) Extremely randomized trees. *Mach Learn* 63:3–42
 27. Grün B, Leisch F (2007) Fitting finite mixtures of generalized linear regressions in R. *Comput Stat Data Anal* 51:5247–5252
 28. Honarvar A, Rozhon J, Millington D, Walden T, Murillo CA, Walden Z (2011) Economic impacts of new oil sands projects in Alberta (2010–2035). Canadian Energy Research Institute, Calgary, Alberta
 29. Huang G, Zhu Q, Siew C-K (2006) Extreme learning machine: theory and applications. *Neurocomputing* 70:489–501
 30. Huo W, Li W, Zhang Z, Sun C, Zhou F, Gong G (2021) Performance prediction of proton-exchange membrane fuel cell based on convolutional neural network and random forest feature selection. *Energy Convers Manage* 243:114367
 31. Janizadeh S, Vafakhah M, Kapelan Z, Mobarghaee Dinan N (2022) Hybrid XGboost model with various Bayesian hyperparameter optimization algorithms for flood hazard susceptibility modeling. *Geocarto Int* 37:8273–8292
 32. Leisch F (2004) FlexMix: a general framework for finite mixture models and latent class regression in R. *J Stat Softw* 11:1–18
 33. Liu B, Zhou Y, Wang D, Dun Y (2023) An unsupervised chatter detection method based on AE and merging GMM and K-means. *Mech Syst Signal Process* 186:109861
 34. Liu X, Tang H, Ding Y, Yan D (2022) Investigating the performance of machine learning models combined with different feature selection methods to estimate the energy consumption of buildings. *Energy Build* 273:112408
 35. Liu Y, Luo H, Zhao B, Xiaoyong Z, Han Z (2018) Short-term power load forecasting based on clustering and XGBoost method, 2018 IEEE 9th International Conference on Software Engineering and Service Science (ICSESS), pp. 536–539
 36. Liu Y, Wu J, Wang Z, Lu X-G, Avdeev M, Shi S, Wang C, Yu T (2020) Predicting creep rupture life of Ni-based single crystal superalloys using divide-and-conquer approach based machine learning. *Acta Mater* 195:454–467
 37. Lu Y, Tian Z, Peng P, Niu J, Li W, Zhang H (2019) GMM clustering for heating load patterns in-depth identification and prediction model accuracy improvement of district heating system. *Energy Build* 190:49–60
 38. Ma S, Fan C, Liu WV (2024) Effects of site operating conditions on real site TKPH (tonne-kilometer-per-hour) of ultra-large off-the-road tires. Part D: *Journal of Automobile Engineering* 238:2227–2239
 39. Ma S, Wu L, Liu WV (2023) Numerical investigation of temperatures in ultra-large off-the-road tires under operating conditions at mine sites. *J Thermal Sci Eng Appl* 15:021010
 40. McLachlan GJ, Lee SX, Rathnayake SI (2019) Finite mixture models. *Annual Rev Stat Application* 6:355–378
 41. Mehrjou A, Hosseini R, Nadjar Araabi B (2016) Improved Bayesian information criterion for mixture model selection. *Pattern Recogn Lett* 69:22–27
 42. MEP (2023) Current and historical Alberta weather station data viewer. Ministry of Environment and Parks (MEP), Government of Alberta, Edmonton, Canada
 43. Ni L, Wang D, Wu J, Wang Y, Tao Y, Zhang J, Liu J (2020) Streamflow forecasting using extreme gradient boosting model coupled with Gaussian mixture model. *J Hydrol* 586:124901
 44. Olu-Ajayi R, Alaka H, Sulaimon I, Balogun H, Wusu G, Yusuf W, Adegoke M (2023) Building energy performance prediction: a reliability analysis and evaluation of feature selection methods. *Expert Syst Appl* 225:120109
 45. Paliwal M, Kumar UA (2011) Assessing the contribution of variables in feed forward neural network. *Appl Soft Comput* 11:3690–3696
 46. Pasch O, Uludag S (2018) Optimization of the load-and-haul operation at an opencast colliery. *J South Afr Inst Min Metall* 118:449–456
 47. Qiu Y, Zhou J, Khandelwal M, Yang H, Yang P, Li C (2022) Performance evaluation of hybrid WOA-XGBoost, GWO-XGBoost and BO-XGBoost models to predict blast-induced ground vibration. *Eng Comput* 38:4145–4162
 48. Saeed U, Jan SU, Lee Y-D, Koo I (2021) Fault diagnosis based on extremely randomized trees in wireless sensor networks. *Reliab Eng Syst Saf* 205:107284
 49. Schexnayder C, Weber SL, Brooks BT (1999) Effect of truck payload weight on production. *J Constr Eng Manag* 125:1–7
 50. Sekhar Roy S, Roy R, Balas VE (2018) Estimating heating load in buildings using multivariate adaptive regression splines, extreme learning machine, a hybrid model of MARS and ELM. *Renew Sustain Energy Rev* 82:4256–4268

51. Sheikhan M, Bejani M, Gharavian D (2013) Modular neural-SVM scheme for speech emotion recognition using ANOVA feature selection method. *Neural Comput Appl* 23:215–227
52. Shimizu N, Kaneko H (2020) Direct inverse analysis based on Gaussian mixture regression for multiple objective variables in material design. *Mater Des* 196:109168
53. Sinaga KP, Yang MS (2020) Unsupervised K-means clustering algorithm. *IEEE Access* 8:80716–80727
54. Song S, Marks E, Pradhananga N (2017) Impact variables of dump truck cycle time for heavy excavation construction projects. *J Construction Eng Project Manage* 7:11–18
55. Sun X, Zhang H, Tian F, Yang L (2018) The use of a machine learning method to predict the real-time link travel time of open-pit trucks. *Math Probl Eng* 2018:4368045
56. Świtafa BM, Guardiani C, Soranzo E, Wu W (2023) Machine learning-aided reliability analysis of rainfall-induced landslide of root-reinforced slopes. *Can Geotech J* 60:12
57. Tibshirani R, Walther G, Hastie T (2001) Estimating the number of clusters in a data set via the gap statistic. *J Royal Stat Soc: Series B (Statistical Methodology)* 63:411–423
58. Ugurlu OF, Fan C, Jiang B, Liu WV (2024) Deep neural network models for improving truck productivity prediction in open-pit mines. *Min Metall Explor* 41:619–636
59. Virupakshappa K, Oruklu E (2019) Unsupervised machine learning for ultrasonic flaw detection using Gaussian mixture modeling, K-means clustering and mean shift clustering, 2019 IEEE International Ultrasonics Symposium (IUS), pp. 647–649
60. Wang Y, Niu L, Zhao L, Wang B, He Z, Zhang H, Chen D, Geng J (2021) Gaussian mixture model deep neural network and its application in porosity prediction of deep carbonate reservoir. *Geophysics* 87:M59–M72
61. Yuan C, Yang H (2019) Research on K-value selection method of K-means clustering algorithm. *J* 2:226–235
62. Zhang Y, Gao S, Cai P, Lei Z, Wang Y (2023) Information entropy-based differential evolution with extremely randomized trees and LightGBM for protein structural class prediction. *Appl Soft Comput* 136:110064
63. Zhao J, Fan C, Huang G, Guo Y, Arachchilage CB, Gupta R, Liu WV (2024) Machine learning-assisted characterization of the thermal conductivity of cement-based grouts for borehole heat exchangers. *Constr Build Mater* 449:138506
64. Zhou H, Deng Z, Xia Y, Fu M (2016) A new sampling method in particle filter based on Pearson correlation coefficient. *Neurocomputing* 216:208–215
65. Zhou H, Zhang J, Zhou Y, Guo X, Ma Y (2021) A feature selection algorithm of decision tree based on feature weight. *Expert Syst Appl* 164:113842
66. Zhou J, Li X, Mitri Hani S (2016) Classification of rockburst in underground projects: comparison of ten supervised learning methods. *J Comput Civ Eng* 30:04016003

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.